

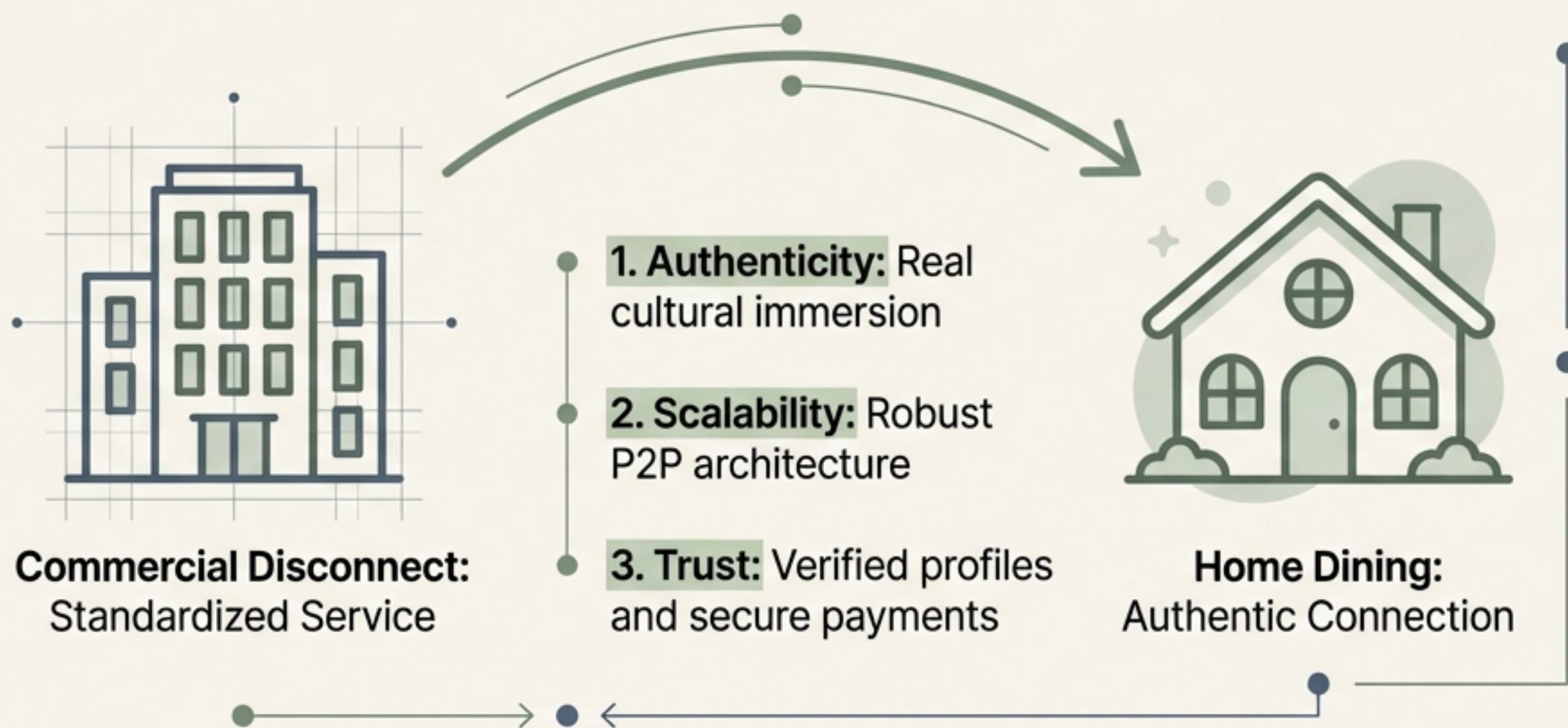


Home-Based Dining Platform

Connecting Guests with Authentic Culinary Experiences

UNIVERSITY PROJECT: SYSTEM DESIGN & IMPLEMENTATION

Bridging the Gap: The Case for Peer-to-Peer Dining



ACADEMIC CONTEXT

Sharing Economy

Principles: Leveraging resource optimization and decentralization (Belk, 2014)

Social Dining:

Addressing authenticity, privacy, and host-guest expectations (Davies et al., 2022)

Competitive Landscape & Differentiation

EatWith

Strong brand, verified hosts.

Limitation: Limited regional coverage, higher fees.

VizEat

European focus, cultural immersion.

Limitation: Merged into EatWith, less independence.

HomeFood (Italy)

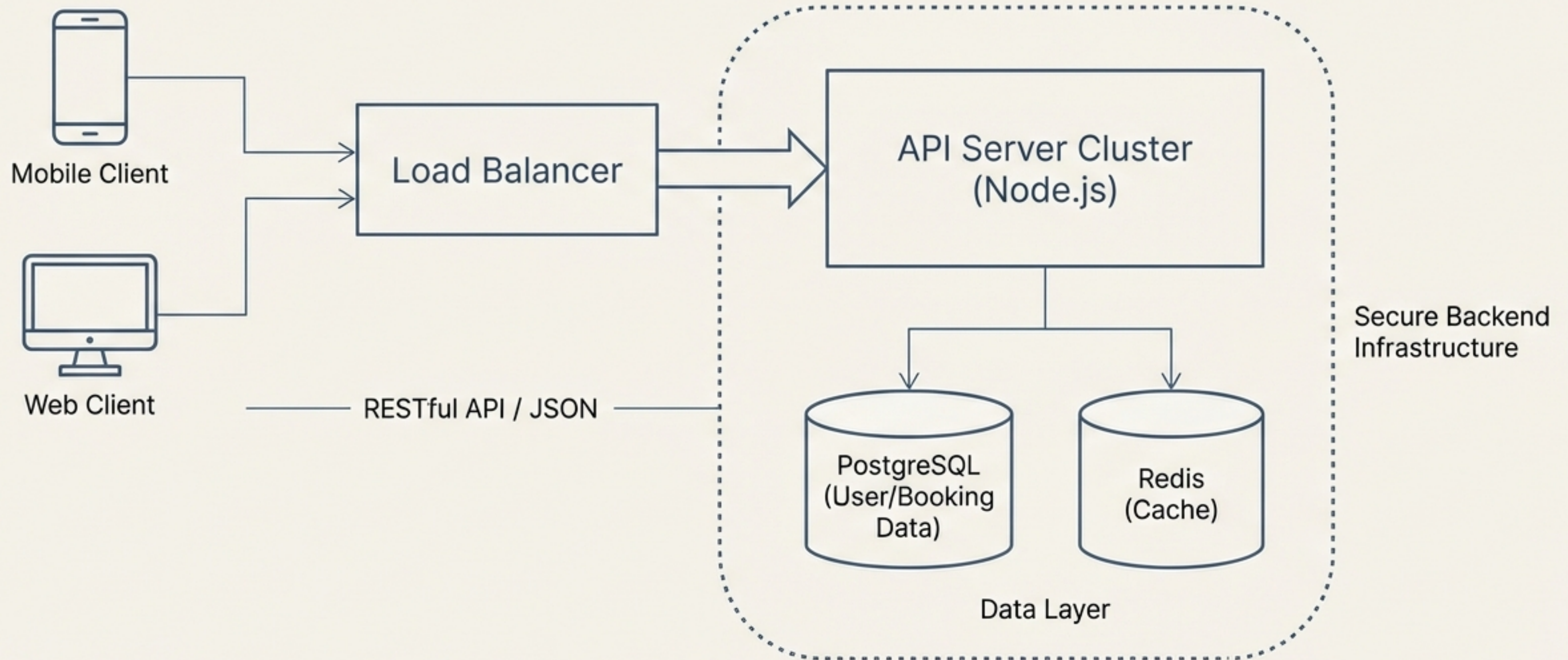
High authenticity.

Limitation: Limited technology and scalability.

Our Strategic Focus (The Gap)

- **Local Chef Support:**
Tools for independent business management.
- **Rich Communication:**
Enhanced rapport-building messaging.

High-Level System Architecture



Technology Stack & Engineering Standards

Frontend



Responsive, component-based UI with fast, consistent styling.

Backend



Unified JavaScript environment allowing code reuse and scalability (Lit Review 6.2)

Data & Infra

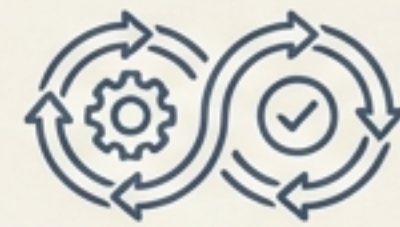


Reliable transactional management, high-performance caching, and containerized deployment.

Principles



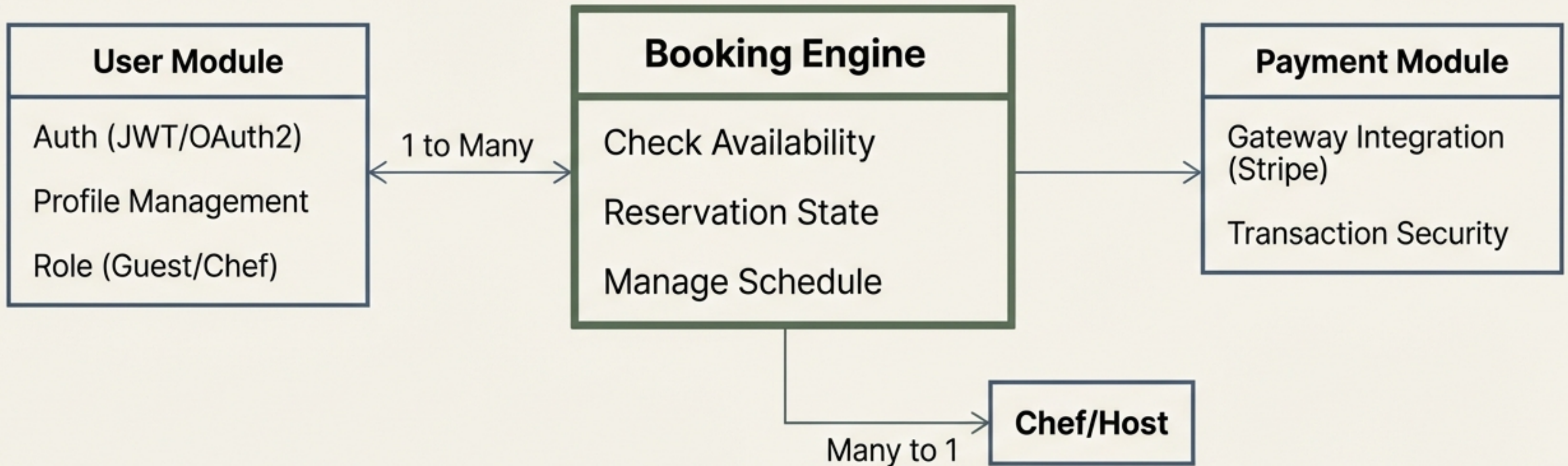
SOLID Architecture



CI/CD Pipeline

Modular design with automated testing to maintain reliability.

Detailed Design: Modular Decomposition



The Booking Engine orchestrates the state between Users, Hosts, and Financial Transactions.

Technical Trade-off: Database Selection

Option A: MongoDB (NoSQL)

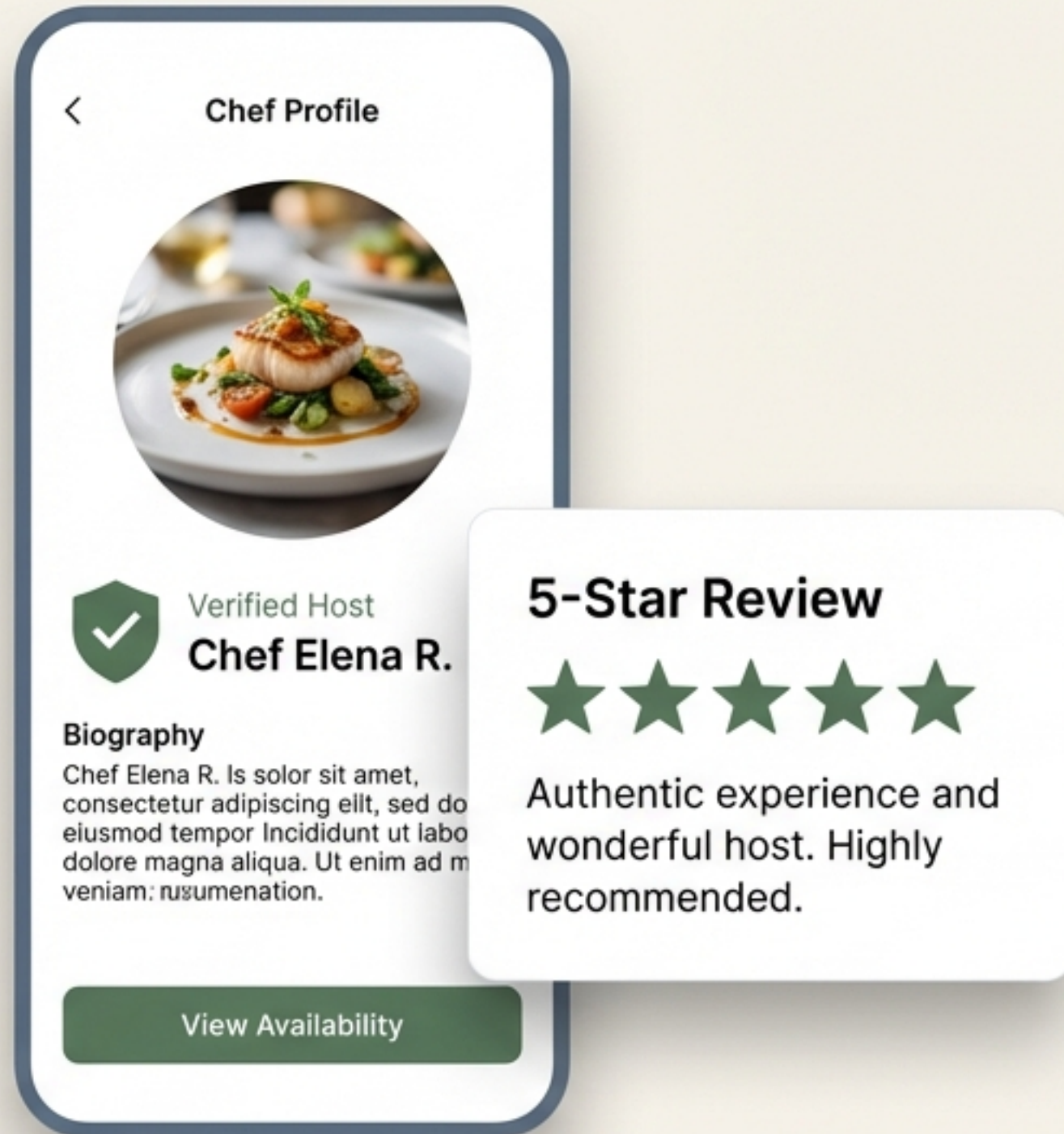
- Pro: Flexible schema for menus.
- Pro: Horizontal scalability.
- Con: No complex joins.
- Con: Weak data integrity constraints.

Option B: PostgreSQL (Relational) - **SELECTED**



- **Why:** Strict ACID compliance required for financial transactions.
- **Rationale:** Relational structure handles Booking-to-User integrity better than NoSQL.
- **Result:** Zero data-loss architecture.

User Experience & Trust Engineering



Trust Features

1. Identity Verification (Badges to reduce risk perception).
2. Social Proof (Transparent rating systems).
3. Visual Assurance (Quality-validated galleries).

Next Steps: Finalize API Specifications and commence Sprint 1.